

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 8: Appendix

by

TAEHONG KWON

Student number: 28576810

submitted in accordance with the requirements for the degree of

DOCTOR OF PHILOSOPHY

in the subject

COMPUTER SCIENCE

at the

UNIVERSITY OF SOUTH AFRICA

SUPERVISOR: Prof. Ernest Mnkandla

CO-SUPERVISOR: Prof. Donatien Koulla Moulla

CO-SUPERVISOR: Dr. David Sena Attipoe

October 2026

8 Appendix

8.1 Evaluation Pipeline and Reproducibility

This appendix documents the evaluation pipeline behind Chapter 4: its stages, the extraction design, the default parameters, the proxy formulas and a glossary.

8.1.1 Pipeline stages

The evaluation has four stages. The first three run on Arbitrum One over the observation period 2025-12-01 to 2026-05-31 (UTC, inclusive at both ends); the fourth adds the months from June to August 2026.

1. Reputation graphs and scores: Fix the matched cohort ($n = 5,521$) from GMX V2 data before any approval or transfer extraction. Fetch the cohort's Approval and Transfer logs in monthly chunks, with a dry run before each query and an abort limit of 150 GiB. Build and score the latest-allowance graph (EndorseRank) and the transfer graph (AWP), then the two raw-amount layers (C-PR at $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$ and S-PR), all with the solver settings of Section 3.1.
2. Checks, alignment and benchmarks: Compute the checks of Section 3.9 and report Spearman ρ and Kendall τ for every score with 95% paired-bootstrap intervals (400 resamples, seed 42), together with the contrasts of Section 3.10. Time one solver call per method under the protocol of Section 3.11, and run the damping, top-token and sample-definition sweeps.
3. Temporal holdout: On the spender cohort ($n = 1,335$), freeze the scores at 28 February 2026 and count the new approval pairs and new transfer senders that appear in March–May 2026. Report every score and both raw-degree baselines against both labels with 400 paired resamples, along with the contrasts fixed before the coupled run and those computed afterwards (s

`cripts/run_holdout_eval.py--cohortspenders`).

4. Registered replication: With `config/fresh_holdout_2026q3.yaml` and its plan, `docs/fresh_holdout_2026q3_plan.md`, committed, first check the evaluation code against the spring holdout (`scripts/run_fresh_holdout.py--spring-check`). Then extract the June–August logs with the same SQL and wallet filter (`scripts/extract_fresh_window.py--dry-run`, then `--extract`) and evaluate both cohorts on scores frozen at 31 May 2026 (`scripts/run_fresh_holdout.py`). The extraction script will not run before the registration is committed with the status set to registered, and both scripts record the SHA-256 hashes of the two registration files.

The coupling weights, holdout labels and the first rule for RQ5 were written to the configuration and to the plan of 14 September 2026, before any coupled score was computed. On 30 September 2026 two inverse-risk proxies were found to carry the wrong sign and were corrected (Appendix 8.1.5). The same day `scripts/classify_spenders.py` checked on BigQuery which spenders emitted logs or sent transactions in the scoring window, and `scripts/check_spender_code.py` read each spender’s code from an Arbitrum node (`data/2-processed-tables-and-evaluations/spring-holdout-2026-03-to-2026-05/spender_account_types.csv`). On 1 October 2026 the registration gained its two sensitivity analyses (Section 3.7.2); AWP’s value parameter is `reputation.awp_paper` in the configuration. Later that day the status was set to registered (commit 4085ab9 of `phd_works`), the June–August logs were extracted in nine monthly queries billed at about 1.04 TB, and `scripts/run_fresh_holdout.py` evaluated them; `data/3-published-results-for-thesis/registered-replication-2026-06-to-2026-08/` holds the extraction manifest and the summary behind Tables 4.14, 4.15 and 4.17.

The other tables of Chapter 4 come from runs made on 2 October 2026 in a cloud container with four cores of an Intel Xeon processor at 2.8 GHz and 15 GB of memory, under Python 3.11 with NumPy 2.4, SciPy 1.17 and pandas 3.0: `scripts/run_dissertation_eval.py--real--robustness--benchmark-repeats5`; `scripts/run_holdout_eval.py` with `--cohortspenders`

and with `--cohortmatched`; `scripts/run_fresh_posthoc.py`, which scores `EndorseRank` on the cohorts and labels of the registered replication (Table 4.16); and `scripts/run_supplementary_checks.py`, behind Tables 4.5 and 4.12. The same day `scripts/run_fresh_holdout.py --spring-check` was run again on the spring window and reproduced the values it had given on 1 October. The tables were written to `results/tables/` by `scripts/export_latex_results.py` and the figures by `scripts/generate_dissertation_figures.py`. The summaries (`eval_summary.json`, `supplementary_checks.json`, `eval_summary_spenders.json` and `eval_summary_matched_traders.json` in the `spring-holdout` folder, and `fresh_holdout_summary.json` and `posthoc_endorserank.json` in the `registered-replication` folder) are committed under `data/3-published-results-for-thesis/` along with a copy of the configuration in force, so any number in a table can be checked against its summary without re-running the pipeline. The public repository https://github.com/abitocodes/phd_works contains the evaluation source in `2-Dissertation-Draft-Works/wallet-reputation-experiments` (the folder was `A-Skill-Programs/margin_rank` until 1 October 2026 and has that name in commit 4085ab9), the versioned configuration (`config/margin_config.yaml`), the extracted data and the summaries, at revision `[revision-pending-commit]`. The file `data/README.md` in that folder describes every data file.

The implementation is checked offline before any real extraction: synthetic tests run the same scoring and alignment logic on fixture graphs, with no cloud access. Real extraction needs Google Cloud credentials and a billing project, and its scans are confined to the public Arbitrum One log dataset.

8.1.2 BigQuery extraction design

Each ERC-20 Approval or Transfer event is identified by its standard topic-0 hash. Queries are restricted to logs whose owner/spender or from/to address is on the evaluation wallet list, which ties scan cost to the size of the cohort instead of the whole chain. Batching by month keeps every query under the operational byte cap. Appendix 8.2 shows representative SQL patterns.

For auditing, the study’s acquisition log retains the extraction records (bytes processed, row counts, timestamps). This research reports only the resulting cohort sizes and edge counts, not the operational logs.

A second-tier extraction had been planned for the 93,559 supplemental addresses of the expanded wallet pool ($N = 99,080$; Section 3.12.1). For the six monthly approval and transfer queries over those addresses, the dry-run estimate came to more than 1.3 TiB, and the download stage failed repeatedly on the storage-backed result path. The extraction was therefore abandoned within the study period. As a result, the supplemental addresses enter the expanded-pool experiments only as sampled addresses. They have no approval or transfer edges of their own, and the solver graph takes its nodes from the endpoints of the retained edges, so they add no nodes either; that is why the full-pool row of Table 4.2 reproduces the matched-cohort graph. Chapter 6 lists the completion of that extraction as future work.

8.1.3 Scaling tables

Table 8.1 reports runtime scaling within the cohort for $n \leq 5,521$, measured under the timing protocol of Section 3.11. Its last row is the same measurement as the one in Table 4.1. Every in-cohort stage is a subset of the wallets whose edges were extracted, so $|V|$ and $|E|$ both grow with n . In the expanded-pool stages they grow as well, but only up to the size of the matched-cohort graph.

Table 8.1: In-cohort runtime scaling (SHA256 seed benchmark-scale-v1; $n \leq 5,521$ matched GMX cohort). Here both $|V|$ and $|E|$ grow with n because every stage is a subset of wallets whose edges were extracted. Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs.

n	ER (s)	AWP (s)	Speedup	ER $ V $	ER $ E $	AWP $ V $	AWP $ E $
1,000	0.096	0.821	$8.6\times$	1,302	2,584	6,463	23,058
2,000	0.185	1.755	$9.5\times$	2,362	5,147	11,849	47,031
4,000	0.392	3.645	$9.3\times$	4,645	10,557	22,591	97,105
5,521	0.547	5.029	$9.2\times$	6,442	14,727	30,791	137,087

8.1.4 Credit-delegation extract (deferred)

Aave V3 records credit delegation in `BorrowAllowanceDelegated` events, a second on-chain form of delegated spending power. This study did not extract them, so the delegation-based row of the archived matrix (Appendix 8.4) has no edges; adding these events to the allowance graph is future work.

8.1.5 Supplementary alignment summary table

The pipeline computes six proxy families, and Table 8.2 lists the family-mean τ of every score on each. Among them are the liquidation, inverse-risk and GMX trading-success families, which were withdrawn from the primary analysis along with the GF-PR reference (Section 2.8). Until 30 September 2026 the pipeline negated two of the three inverse-risk proxies, the summed loss and the worst close, so both ranked the largest losses highest. They are now oriented like every other proxy, a value nearer zero meaning smaller losses, and the table shows the corrected values; the correction changes no other entry. Every score is negative on this family, which fits the summed loss and the worst close growing with how much a wallet trades (Section 4.3). The transfer, allowance and Sybil-stability columns repeat Table 4.3 without intervals.

Table 8.2: Same-window mean Kendall τ by proxy family for every score ($n = 5,521$). The first three columns repeat Table 4.3 without intervals; the liquidation, inverse-risk and GMX trading-success families are archived.

Method	Transfer	Allowance	Sybil	Liq.	Inv.risk	GMX
EndorseRank	0.330	0.375	0.231	0.093	-0.055	0.096
AWP	0.612	-0.024	0.278	0.141	-0.127	0.236
C-PR ($\lambda = 1$)	0.333	0.355	0.233	0.093	-0.056	0.096
C-PR ($\lambda = 0$)	0.534	-0.030	0.286	0.113	-0.094	0.179
C-PR ($\lambda = 0.25$)	0.531	-0.009	0.288	0.114	-0.093	0.176
C-PR ($\lambda = 0.5$)	0.527	-0.001	0.290	0.113	-0.091	0.171
C-PR ($\lambda = 0.75$)	0.519	0.006	0.292	0.110	-0.086	0.164
S-PR	0.494	0.116	0.220	0.119	-0.099	0.181

8.1.6 Parameter summary

Table 8.3: Default parameters used in the primary evaluation.

Parameter	Value	Role
PageRank damping d	0.85	Default teleport mixture
Convergence tolerance ε	10^{-8}	Power-iteration stop
Max iterations $I_{\max}$	300	Power-iteration cap
Logistic k	0.01	Decay slope (AWP, EndorseRank, transfer layer)
Logistic t_0	180 days	Decay midpoint (AWP, EndorseRank, transfer layer)
Value transform b	1 (raw base units)	Value weight $V(z)$ of AWP and EndorseRank
Restart distribution	$\propto X_u$; uniform	AWP; EndorseRank, C-PR
Coupling weight λ	0.5 (primary); 0.25, 0.75	C-PR allowance-layer weight, fixed before the run
S-PR restart vector	allowance walk, normalized, floor 0	Endorsement-seeded teleportation
Holdout freeze / outcome window	2026-02-28 / 2026-03-01 – 2026-05-31	Spender cohort $n = 1,335$
Registered replication freeze / labels	2026-05-31 / 2026-06-01 – 2026-08-31	Spender and trader cohorts; margin 0.02
Observation window	2025-12-01 – 2026-05-31	UTC inclusive
Bootstrap resamples / seed	400 / 42	Paired wallet bootstrap, percentile 95% CI
Timed repeats / warm-up	5 / 1	Runtime protocol
In-cohort subsample seed	benchmark-scale-v1	Deterministic SHA256 seed
Expanded pool subsample seed	benchmark-tier2-v1	Deterministic SHA256 seed

8.1.7 Proxy operational definitions (reference)

The formulas below are those of Chapter 3.

- Transfer: $\text{in_degree}(w) = |\{u : \exists u \rightarrow w\}|$; $\text{in_value}(w) = \sum_{e:u \rightarrow w} \text{value}(e)$.
- Allowance: $\text{in_approve_degree}(w) = |\{o : \exists o \rightarrow w\}|$; $\text{in_approve_value}(w) = \sum_{o, \text{token}} a^*(o, w, \text{token})$, where w is the spender.
- Sybil-adjusted stability: $\text{inbound_counterparty_ratio}(w) = |\{u : u \rightarrow w\}| / |T^{\text{in}}(w)|$ (self-transfers excluded); tenure in days; active months.

8.1.8 Glossary

Activeness AWP’s restart weight of an address: over the addresses it has sent to, the sum of the time weights of its latest transfer to each.

Allowance An ERC-20 authorization that lets a spender move tokens on an owner’s behalf.

Alignment Rank agreement between reputation scores and a validation proxy, measured by Spearman’s ρ on mid-ranks and by Kendall’s τ ; every reported τ is the tie-corrected τ_b .

Construct overlap A ranking method and a proxy sharing a data source, which inflates alignment.

C-PR end points C-PR at $\lambda = 1$ and at $\lambda = 0$, which walk the raw-amount allowance layer or transfer layer alone with uniform restarts. The registration documents call them EndorseRank and AWP.

Endorsement graph Directed graph of latest allowances, used as the input to EndorseRank and as the allowance layer of C-PR.

Matched cohort Primary sample of $n = 5,521$ GMX V2 wallets on Arbitrum One with at least three closes between 2025-12-01 and 2026-05-31, fixed from GMX data before any approval or transfer extraction. Membership does not depend on graph connectivity: 966 of these wallets touch no allowance edge and 381 touch no transfer edge.

Expanded wallet pool A larger address set ($N = 99,080$) for testing runtime sensitivity to the number of sampled wallets; its supplemental addresses have no extracted edges, so they add no nodes to the solver graph.

Paired bootstrap Resampling of wallets with replacement through a single index matrix shared by every coefficient, so that differences between coefficients can be tested on the same re-samples.

PageRank Spectral ranking of nodes by the stationary distribution of a damped random walk.

Proxy Observable wallet metric for checking rankings against domain intuition.

Spender cohort Holdout sample of $n = 1,335$ wallets: every spender that holds at least one positive latest allowance at t_1 (28 February 2026, 23:59:59 UTC) in the extracted approval data.

Teleport The jump that PageRank makes with probability $1 - d$, to a node drawn from the restart distribution: uniform for EndorseRank and C-PR, weighted by sending activity for AWP.

8.2 SQL Templates, Configuration, and Extended Notes

To aid reproduction, this appendix gathers representative SQL patterns, configuration keys, and extended methodological notes.

8.2.1 Approval log extraction pattern

An ERC-20 Approval event is identified by its topic 0, which is the keccak256 hash of Approval(address,address,uint256). For one month, a simplified query is:

```
SELECT
    block_timestamp,
    block_number,
    transaction_hash,
    log_index,
    address AS token,
    topics[SAFE_OFFSET(1)] AS owner_topic,
    topics[SAFE_OFFSET(2)] AS spender_topic,
    data AS value_data
FROM `bigquery-public-data.google_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
    AND block_timestamp <  TIMESTAMP('2026-01-01')
    AND topics[SAFE_OFFSET(0)] =
    ↪ '0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925'
    AND (
        LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
        ↪ UNNEST(@wallet_list)
```

```

    OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
    ↪ UNNEST(@wallet_list)
);

```

With rows filtered against @wallet_list, scan volume is proportional to the cohort instead of the full chain. Monthly batches keep every query below the per-query byte caps documented in the study's acquisition record.

8.2.2 Transfer log extraction pattern

Transfer events are selected by the topic 0 of Transfer(address,address,uint256):

```

SELECT
    block_timestamp,
    block_number,
    transaction_hash,
    log_index,
    address AS token,
    topics[SAFE_OFFSET(1)] AS from_topic,
    topics[SAFE_OFFSET(2)] AS to_topic,
    data AS value_data
FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
    AND block_timestamp <  TIMESTAMP('2026-01-01')
    AND topics[SAFE_OFFSET(0)] =
    ↪ '0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'
AND (
    LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
    ↪ UNNEST(@wallet_list)
    OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
    ↪ UNNEST(@wallet_list)

```

);

Before PageRank, post-processing keeps, for each (token, owner, spender) triple, the value of the last Approval event in (block_number, log_index) order and drops zero (revoked) values. It then weights each latest approval and each transfer by its time weight and bounded value weight, builds the raw-amount layers of C-PR beside them, and keeps every edge with at least one endpoint in the matched cohort. Amounts stay in raw token base units, the uint256 value cast to a float, and no decimal scaling is applied before the value weight or, in the C-PR layers, before allowances or transfer values are summed across tokens.

8.2.3 Configuration keys

All of the pipeline’s parameters are kept in a single versioned configuration file, whose principal keys include:

- start and end timestamps of the observation window;
- the PageRank damping factor, tolerance, and maximum number of iterations;
- logistic decay parameters k and t_0 and the value-weight parameter b ;
- subsample seeds and the number of benchmark repetitions;
- the locations to which intermediate artifacts and table fragments are written (results/tables/).

Changing the damping factor or the decay parameter requires a rerun of the entire evaluation, including the robustness sweeps and the table export. All figures must then be regenerated from the same artifacts so that the reported numbers match the pipeline’s machine-readable summary.

8.2.4 Complexity notes

Write $m = |E|$. Building the latest-allowance edges sorts the R_a approval rows in (block_number, log_index) order, in $O(R_a \log R_a)$ time, and the transfer edges take $O(R_t)$ over the transfer rows. PageRank over I iterations costs $O(m + mI)$, the first term for building the sparse operator,

which dominates the measured calls (Appendix 8.3.3). The edge lists and score vectors account for most of the memory, so the peak-memory ratio in Table 4.1 is of the order of the edge ratio.

8.2.5 Conventions for the result tables

The tables of Chapter 4 follow these conventions:

1. Every τ is Kendall's τ_b on raw scores and proxies, reported with its 95% paired-bootstrap interval. Differences between coefficients come from the paired contrast tables, not from subtracting point estimates.
2. On the matched cohort, EndorseRank's coefficients are capped by its ties (Section 4.4), so they are compared with other coefficients only through the contrasts and with that cap in mind.
3. A runtime is the time of one PageRank call on edge lists built in advance, covering matrix assembly and power iteration but not extraction or preprocessing. The same graph under the same settings shows the same numbers in every table (Section 3.11).
4. The sample-definition sweep (Table 4.8) is a sensitivity analysis, and its absolute values are not compared with the matched-cohort tables.

8.2.6 Archived extended baselines

The archived matrices, and the reasons they stay out of the main comparison, are described in Appendix 8.4.

8.2.7 Limitations

The SQL above was run only on Arbitrum One. All reported numbers come from the pipeline's machine-readable summaries for the matched cohort ($n = 5,521$), the pool subsamples, the spender cohort ($n = 1,335$) and the exploratory trader run; the registered June–August results come from the summary of that run (Section 4.6.2).

8.3 Elementary Properties of PageRank and Rank Correlation

Chapters 2–4 rely on the elementary properties summarized here. The material is standard: the PageRank properties follow Page et al.^[15] and Langville and Meyer^[23], and the rank correlations Spearman^[20] and Kendall^[21].

8.3.1 Existence and uniqueness of PageRank

Take P to be a row-stochastic matrix on $|V|$ nodes, possibly after the dangling-node repair of Equation (2.1). For $d \in (0, 1)$ the Google matrix of Equation (2.2),

$$G_{\text{PR}} = dP^\top + \frac{1-d}{|V|}\mathbf{1}\mathbf{1}^\top,$$

is positive and column-stochastic. By the Perron–Frobenius theorem for positive matrices, G_{PR} has a single eigenvalue of modulus one, and the corresponding eigenvector $\mathbf{r}$ is strictly positive; we can scale it so that $\sum_i r_i = 1$. It follows that, once the damping and the dangling-node rules are fixed, every directed graph has unique PageRank scores^[23].

Two restart vectors in this thesis are not uniform. AWP’s, proportional to activeness, is zero at every address that has sent nothing, and S-PR’s, the normalized score of the allowance walk, is zero at every transfer-layer wallet without an allowance score, so neither Google matrix is positive. The fixed point is still unique: for two probability vectors the restart term cancels in their difference, and the operator multiplies their $\|\cdot\|_1$ distance by at most d . In general it is not strictly positive, however, because a wallet with no restart mass that no path from a restart wallet reaches keeps a score of zero.

Uniqueness guarantees a well-defined comparison between EndorseRank and AWP. Once its parameters are fixed, each method assigns exactly one score vector to a given edge set, so any difference between $\mathbf{s}_{\text{ER}}$ and $\mathbf{s}_{\text{AWP}}$ must arise from their edges and restart distributions and cannot be traced to multiple solvers.

8.3.2 Continuity in edge weights

For fixed $d \in (0, 1)$, PageRank depends continuously on the entries of P . If $\tilde{P}$ is a second row-stochastic matrix on the same nodes, with PageRank vector $\tilde{\mathbf{r}}$, then

$$\|\mathbf{r} - \tilde{\mathbf{r}}\|_1 \leq \frac{d}{1-d} \max_i \|P_{i\cdot} - \tilde{P}_{i\cdot}\|_1,$$

a bound that does not depend on $|V|$ [23].

Continuity in P is not the same as continuity in the edge weights. A row $P_{i\cdot} = w_{i\cdot}/o_i$ moves continuously with the weights only while the out-strength o_i stays positive. When a node gains its first out-edge or loses its last one, its row jumps between the uniform dangling row and a row concentrated on its out-neighbors, however small the weight involved. In the worked example of Section 2.13, adding an edge $D \rightarrow A$ of weight 10^{-12} moves $\mathbf{r}$ by 0.37 in the $\|\cdot\|_1$ norm and raises r_A from 0.1375 to 0.3202.

The damping sweep is covered by continuity: it keeps P fixed and changes d smoothly, and for $d < 1$ the PageRank vector varies continuously with d . The top-token filter is not covered. It removes edges, which can leave a wallet without out-edges, so the stability of that check rests on the data alone (Table 4.7). Empirically, EndorseRank’s allowance family mean is unchanged in the first three decimal places across $d \in \{0.75, 0.85, 0.95\}$ (Table 4.6); with most of the cohort tied, the ranking could hardly have moved.

8.3.3 Sparsity and per-iteration cost

With $m = |E|$ nonzero edges, the sparse product $P^\top \mathbf{x}$ costs $O(m)$ arithmetic operations, and building the sparse operator from the edge list is also $O(m)$, so power iteration over I iterations is $O(m + mI)$. On the matched cohort the transfer graph has about 9.3 times as many edges as the endorsement graph, so even at identical I the transfer method would run roughly an order of magnitude longer. In the measured calls the assembly of the operator takes most of the time and the iterations little (Section 4.2), so the measured time ratio of 9.2 tracks the edge ratio.

8.3.4 Properties of Spearman’s ρ

Spearman’s ρ is Pearson’s correlation computed on ranks. It lies in $[-1, 1]$ and does not change under strictly increasing transformations of either variable. A value of $\rho = 1$ means perfect monotone agreement, and $\rho = -1$ perfect reverse agreement. Tied values receive mid-ranks, and the evaluation pipeline relies on standard tie-aware implementations. Because reputation scores and proxies are continuous and heavy-tailed, rank-based association suits them better than Pearson correlation on the raw values. The AWP baseline paper likewise evaluates its ranking with rank statistics ^[3].

8.3.5 Properties of Kendall’s τ

Kendall’s τ compares the numbers of concordant and discordant pairs, n_c and n_d . Without tie correction, $\tau_a = (n_c - n_d)/n_0$ with $n_0 = n(n - 1)/2$ is the probability that two randomly drawn wallets are ordered the same way by both rankings minus the probability that they are ordered oppositely ^[21]. Every reported τ is the tie-corrected τ_b of Equation (2.6). The two coincide without ties, and here ties are heavy. EndorseRank takes 99 distinct values on the matched cohort, so 67.3% of all wallet pairs are tied on it, against 0.5% on AWP, and for EndorseRank against AWP the reported $\tau_b = 0.360$ corresponds to $\tau_a = 0.206$. Ties also cap τ_b . EndorseRank’s largest attainable value is about 0.38 against the allowance proxies, which are positive for exactly the 131 wallets it separates, and about 0.57 against a proxy without ties, so its 0.375 on the allowance family (Table 4.3) is nearly complete agreement on the pairs the rankings separate and cannot be compared directly with a coefficient that has a different ceiling.

When n is large, τ and ρ usually agree in sign, but they differ in magnitude. Across the pairs of EndorseRank or AWP with a proxy where $|\tau| > 0.02$, the ratio ρ/τ lies between 1.01 and 1.49, and ρ is always the larger of the two. We report both, as the AWP baseline does ^[3]. Our main summary statistic is the family-mean τ with a paired-bootstrap interval ^[34].

8.3.6 Construct validity note

Cronbach and Meehl ^[22] distinguish criterion-related validity from construct validity. Since no definitive gold standard default label exists, this research takes a construct-oriented approach. Each score is tested against validation checks that embody its conceptual definition (e.g. receiving allowances for EndorseRank; receiving inbound transfers for AWP), and also against a holdout period with two labels, new approval pairs and new transfer senders. High within-construct τ supports the interpretation. A low cross-construct τ , by contrast, does not automatically signal failure; it may indicate discriminant validity.

8.3.7 Where each primary quantity is reported

For each primary quantity, the index below names the table where it is established.

Quantity	Reported in
Matched-cohort size, $ V $, $ E $, runtime, SD, memory, iterations (four methods)	Table 4.1
In-cohort scaling stages	Table 8.1
Pool subsample stages ($N = 99,080$)	Table 4.2
Family-mean τ with 95% CI, every score	Table 4.3
Paired $\Delta\tau$ contrasts of C-PR and S-PR	Table 4.4
Inter-method τ ; wallets outside a graph as isolated nodes	Table 4.5
Damping, top-token, sample-definition sweeps	Tables 4.6, 4.7, 4.8
Spender holdout ($n = 1,335$)	Tables 4.10, 4.11 and 4.12
Exploratory trader run	Table 4.13
Registered June–August replication	Section 4.6.2
Comparisons between EndorseRank and AWP, per-proxy results	Appendix 8.5
Observation window	2025-12-01 – 2026-05-31 (Table 8.3)

Each quantity listed appears both in the evaluation summary and in a table of Chapter 4 or of the appendix.

8.4 Archived Extended Baselines

Further PageRank variants were scored on the same matched cohort and proxy families. They are not among the scores evaluated in Chapter 4.

8.4.1 Seven-method alignment matrix

The evaluation archive keeps the seven-method matrices (LP-PR, CW-AWP, LF-PR, and RiskProp-style variants), which are not drawn here. Their rows were computed with the two raw-amount layers of C-PR walked alone as the allowance and transfer references. Among these methods, the allowance walk leads on the allowance family, and the transfer walk, together with its closely related variants, leads on transfer.

The archive matrices show the following patterns:

- Methods that use transfer-like edges cluster around the transfer walk on the transfer and Sybil families.
- When a method has very few edges, its rankings can be unstable or degenerate, and a low runtime may simply reflect a tiny m .
- RiskProp-style rows that match the transfer walk numerically indicate shared transfer structure; they do not constitute an independent discovery.

8.4.2 Six-method Aave-oriented matrix

An Aave-oriented preset is archived as well. Within the study window, several of the lending-event graphs on Arbitrum are sparse, and the delegation-based rows may be degenerate when BorrowAllowanceDelegated events were not extracted (Appendix 8.1).

That preset remains in the archive and is not drawn here.

Although these matrices motivate future work on credit-delegation edges and cross-protocol lending labels, they leave Claims C1–C5 unchanged.

8.4.3 Why these variants are archived

Three reasons explain why the scores evaluated are EndorseRank and AWP, with C-PR and S-PR built from the same two layers:

1. Allowance-based endorsement is the new construct, and evaluating it beside the established transfer method ^[3] isolates the research question.
2. Variants that share transfer edges with AWP contribute extra rows but no independent construct.
3. The outcome-native GF-PR reference from the research proposal is derived from the same GMX events as the trading-success proxies, so any alignment between the two counts as construct overlap and not as validation (Section 2.8). The GF-PR three-method matrix is archived alongside the seven-method and six-Aave presets.

8.4.4 Reproducing archive tables

Running the evaluation pipeline's table export with the seven-method and six-Aave presets regenerates the archive tables. CSV copies are written out together with the table fragments.

8.4.5 Notes on archived method identifiers

The archive tables and CSV exports use the identifiers below.

LP-PR The raw-amount transfer layer, with AWP's time decay, with each edge weight multiplied by one minus the GMX liquidation rate of its sender and one minus that of its receiver. It tracks the transfer walk closely on the transfer families. Its weights use the same liquidation outcomes as the liquidation proxies, so it shares the circularity of GF-PR.

CW-AWP The raw-amount transfer layer restricted to transfers of a fixed list of collateral tokens. Its transfer τ is slightly lower than that of the full layer.

LF-PR A lending-flow graph from Aave events, with borrows as edges from the pool to the user and repayments and liquidations as edges from the user to the pool, all under AWP’s time decay. Few cohort wallets used Aave on Arbitrum in the window, so the graph has very few edges and near-zero runtimes, and its rankings cannot be compared with the transfer walks.

RiskProp The raw-amount transfer layer with each edge weight multiplied by one minus the GMX liquidation rate of its sender. It borrows the idea of propagating risk from Lin et al. ^[12] but is not their method. On this cohort it mirrors the transfer walk numerically on several families, so the liquidation weighting changed little.

LiqCall-PR / Borrow-PR / Repay-PR Aave-oriented wallet-to-wallet graphs derived from liquidation calls, borrows, and repays, where applicable. On Arbitrum in the study window they have only tens to hundreds of edges, which leaves alignment weak and runtimes negligible.

Delegation-PR Intended to use credit-delegation allowances. With zero delegation edges extracted, its scores are degenerate and its τ cannot be interpreted.

No archive row is a peer-reviewed baseline.

8.4.6 Criteria for the scores evaluated

Chapters 4–5 concentrate on EndorseRank, AWP and the walks that couple their two layers; the other variants fall short of the following criteria:

1. Construct independence: a baseline should implement different edge semantics instead of merely reweighting transfers.
2. Data adequacy: there should be enough edges for PageRank to be stable on the matched cohort.
3. Scope: the contribution is allowance-based endorsement; it is not meant to survey every possible PageRank variant.

By these criteria, EndorseRank (allowance) and AWP (transfer), with the walks that couple their layers, make up a minimal sufficient set. The seven-method and six-Aave matrices remain archived and do not enter into Claims C1–C5.

8.5 EndorseRank and AWP Side by Side

Chapter 4 reports every score of the thesis together and reads each PageRank against the raw degree it smooths. This appendix collects the direct comparisons between the two single-layer scores, EndorseRank and AWP: their family means and paired differences in the same window, their per-proxy coefficients, and their paired differences out of window. None of these comparisons was fixed in advance, and none carries a claim of the thesis. The two scores walk on different edges and are built for different constructs, so on each check the score ahead is, almost every time, the one whose edges the check counts. Their coverage differs as well. No allowance edge touches 966 of the matched wallets and no transfer edge touches 381, and 318 of the 1,335 holdout spenders have no transfer edge before the freeze. EndorseRank’s ties decide most of its same-window ordering (Section 4.4). In both holdout windows each of the two falls below the raw degree it smooths (Section 4.6), so a difference between them does not show that either adds anything to its degree.

8.5.1 Same window

Table 4.3 gives the family means of the two scores with their intervals, and the last row of Table 4.5 their agreement. The two patterns cross. AWP tracks the transfer family most closely (0.612) and is slightly negative on the allowance family, while EndorseRank tracks the allowance family most closely (0.375) and the transfer family moderately (0.330). Table 8.4 tests five paired differences on the same wallet resamples (Section 3.10), numbered i to v in the order of its rows.

In three of the five contrasts the interval excludes zero. AWP’s transfer alignment is higher than EndorseRank’s (+0.282), EndorseRank’s allowance alignment is higher than AWP’s (+0.398), and AWP leads EndorseRank on the Sybil-stability family (+0.047). The remaining two compare En-

Table 8.4: Paired bootstrap tests of differences between Kendall τ coefficients (matched cohort, $n = 5,521$). Family entries use the family-mean τ ; the same wallet resamples are used for a and b , so the interval is on the paired difference. An interval excluding 0 indicates a difference not attributable to sampling variation at the 5% level. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
EndorseRank allowance minus EndorseRank transfer	0.375	0.330	+0.045	[-0.001, 0.084]	0.970
AWP transfer minus EndorseRank transfer	0.612	0.330	+0.282	[0.263, 0.299]	1.000
EndorseRank allowance minus AWP allowance	0.375	-0.024	+0.398	[0.356, 0.432]	1.000
AWP sybil-stability minus EndorseRank sybil-stability	0.278	0.231	+0.047	[0.031, 0.062]	1.000
EndorseRank in-approve degree minus EndorseRank in-degree	0.376	0.341	+0.035	[-0.015, 0.079]	0.925

dorseRank with itself, and both are null: its allowance alignment is not shown to be higher than its transfer alignment (contrast i, +0.045, interval $[-0.001, 0.084]$), and its in-approve-degree alignment is not shown to be higher than its in-degree alignment (contrast v, +0.035, $[-0.015, 0.079]$). These two nulls cannot be read as evidence either way. EndorseRank’s ties cap τ_b at about 0.38 for the allowance proxies and at about 0.57 for a proxy without ties (Section 4.3), so the coefficients are not on one scale. With the wallets outside the allowance graph kept as isolated nodes instead of being scored zero, contrasts i and v become +1.061 and +1.090, with intervals far above zero. The opposite signs of contrasts ii and iii follow from the constructs. Each score is ahead on the family built from its own edges, and the main proxies of each family include the in-degree that the corresponding PageRank smooths. EndorseRank’s lead on the allowance family rests on the 131 wallets that receive approvals, all of which it lifts above its common value, while AWP spreads those 131 over most of its ranking. AWP’s lead on the transfer family is measured against a transfer coefficient of EndorseRank’s that turns negative once the missing wallets are kept as isolated nodes. Neither difference says which score is the better measure of reputation.

On the staged pool subsamples of Table 4.8, whose values are not comparable with those of the matched cohort, the order of the two scores within each family is the same at every stage: AWP is ahead on the transfer family and EndorseRank on the allowance family.

8.5.2 Proxy detail

Each family mean condenses several per-proxy coefficients. Tables 8.5 to 8.7 report them one proxy at a time for EndorseRank and AWP, as Spearman ρ , Kendall τ and the 95% interval of τ . Before correlation, every proxy is oriented so that a higher value means higher expected reputation, and the transfer proxies are the ones AWP’s source paper used to validate it^[3].

8.5.2.1 Transfer family

Table 8.5: Domain alignment with transfer proxies (inbound in-degree and in-value, the validation proxies used for the AWP baseline). Kendall τ and Spearman ρ between reputation scores and proxy rankings. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
In-degree	0.418	0.341	[0.319, 0.363]
In-value (sum inbound)	0.396	0.319	[0.300, 0.340]
<i>AWP</i>			
In-degree	0.884	0.728	[0.718, 0.737]
In-value (sum inbound)	0.669	0.495	[0.481, 0.510]

In Table 8.5, AWP’s agreement with in-degree is the largest per-proxy Kendall τ on the matched cohort, and its agreement with in-value is much lower, for the reason given in Section 4.3. EndorseRank’s transfer coefficients are moderate (0.341 and 0.319), and their intervals overlap neither zero nor AWP’s intervals. They mainly reflect that wallets in the allowance graph also receive more transfers than wallets outside it (Section 4.4). A protocol that equates reputation with inbound flow can read that flow from the in-degree itself, and out of window AWP ranks new senders below the in-degree (Section 4.6).

8.5.2.2 Allowance family

In Table 8.6, EndorseRank’s two coefficients are almost equal and sit at the ceiling that its ties allow (Section 4.3). AWP’s are slightly negative (-0.024 and -0.023), with intervals that exclude

Table 8.6: Domain alignment with allowance proxies (spender-side in-approve degree and value). CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
In-approve degree	0.380	0.376	[0.344, 0.403]
In-approve value (sum)	0.380	0.373	[0.341, 0.398]
<i>AWP</i>			
In-approve degree	-0.029	-0.024	[-0.043, -0.002]
In-approve value (sum)	-0.029	-0.023	[-0.043, -0.002]

zero: among the cohort’s wallets, transfer centrality does not pick out those that are approved as spenders. The comparison rests on the 131 wallets that receive approvals and is partly built in, since EndorseRank is computed from the same approvals.

8.5.2.3 Sybil-adjusted stability family

Table 8.7: Domain alignment with Sybil-adjusted stability proxies from transfer activity. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
Inbound counterparty ratio	0.091	0.072	[0.045, 0.095]
Transfer tenure (days)	0.363	0.294	[0.274, 0.316]
Active months	0.377	0.327	[0.307, 0.348]
<i>AWP</i>			
Inbound counterparty ratio	-0.269	-0.228	[-0.250, -0.207]
Transfer tenure (days)	0.685	0.507	[0.493, 0.521]
Active months	0.704	0.555	[0.540, 0.569]

Table 8.7 shows AWP ahead of EndorseRank on tenure (0.507 against 0.294) and active months (0.555 against 0.327), as one would expect from a transfer score that rewards activity spread over the window; the paired difference on the family mean excludes zero (Table 8.4, contrast iv). On the inbound-counterparty ratio AWP is clearly negative (-0.228) and EndorseRank weakly positive (0.072). The family describes how much longitudinal transfer information each score carries.

On the archived trading families of Table 8.2, AWP is ahead of EndorseRank on liquidation (0.141 against 0.093) and on trading success (0.236 against 0.096). On the inverse-risk family both are negative, -0.055 for EndorseRank and -0.127 for AWP, and on the non-loss close rate, a share that does not grow with trading volume, the two coefficients are -0.031 and -0.006 .

8.5.3 Out of window

Table 8.8 collects the paired differences between the two scores in both holdout windows. On the full spender cohorts, a is the score built from the edge type that the label counts; on the subset of the second block both rows take EndorseRank as a .

Table 8.8: Paired bootstrap differences between EndorseRank and AWP out of window. None was fixed in advance: the spring rows were computed after the spring labels were known, and EndorseRank was scored on the June–August cohorts after the registered evaluation (Table 4.16). The second block is the subset of Table 4.12. Within each block a and b share the wallet resamples (400, seed 42).

Contrast (a minus b)	Label	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
<i>Spring holdout, spenders ($n = 1,335$)</i>						
EndorseRank minus AWP	new approval pairs	0.394	0.123	+0.271	[0.207, 0.332]	1.000
AWP minus EndorseRank	new transfer senders	0.344	0.283	+0.061	[0.026, 0.098]	1.000
<i>Spring holdout, spenders that had received a transfer before the freeze ($n = 977$)</i>						
EndorseRank minus AWP	new approval pairs	0.340	0.298	+0.042	[0.003, 0.085]	0.978
EndorseRank minus AWP	new transfer senders	0.358	0.388	-0.029	[-0.070, 0.011]	0.085
<i>June–August, spenders ($n = 1,964$)</i>						
EndorseRank minus AWP	new approval pairs	0.333	0.118	+0.215	[0.171, 0.270]	1.000
AWP minus EndorseRank	new transfer senders	0.319	0.255	+0.064	[0.036, 0.088]	1.000
<i>June–August, traders with at least three closes ($n = 1,402$)</i>						
EndorseRank minus AWP	liquidation-free close rate	0.081	0.045	+0.035	[-0.017, 0.087]	0.925

On the spring spenders EndorseRank ranks new approval pairs at $\tau = 0.394$ (interval [0.360, 0.431]) against 0.123 for AWP, a paired difference of +0.271 ([0.207, 0.332]). On new transfer senders the order reverses: AWP reaches 0.344 against 0.283 for EndorseRank, a difference of +0.061 ([0.026, 0.098]). Every label falls after all scored events, so on this cohort each edge type carries information about its own construct that the other lacks. Coverage explains much of the first gap.

AWP scores zero for the 318 spenders without a transfer edge before the freeze, and these gained new approval pairs more often than the rest. On the 977 spenders that had received a transfer, EndorseRank’s lead on new approval pairs shrinks to +0.042 ([0.003, 0.085]), and the two scores are level on new senders (-0.029 , $[-0.070, 0.011]$). The in-approve degree, which uses no transfer information, is also ahead of AWP on new senders (0.404 against 0.344, without a paired test).

In June–August, with EndorseRank scored after the registered evaluation, the pattern repeats. EndorseRank ranks new approval pairs at 0.333 against 0.118 for AWP (+0.215, [0.171, 0.270]), and AWP ranks new senders at 0.319 against 0.255 (+0.064, [0.036, 0.088]). On the liquidation-free close rate of the traders EndorseRank reaches 0.081 and AWP 0.045, a difference that cannot be told from zero (+0.035, $[-0.017, 0.087]$).

On the matched traders of the exploratory spring run (Table 4.13), AWP ranks the number of profitable closes at 0.210 (interval [0.181, 0.238]) against 0.072 ([0.041, 0.111]) for EndorseRank, and the realized gain at 0.191 ([0.159, 0.219]) against 0.048 ([0.017, 0.082]). On the share of closes that were profitable neither score shows anything: 0.016 ($[-0.015, 0.044]$) for AWP and -0.016 ($[-0.053, 0.024]$) for EndorseRank. AWP’s lead is on counts and sums, which grow with how much a wallet trades, and the transfer in-degree does about as well on the same two labels.

8.6 Claim-to-Evidence Map

This appendix maps each hypothesis, research question, and empirical claim to the relevant evidence in Chapter 4.

8.6.1 H1 / RQ1: Rank divergence

- Statement: on the same cohort, EndorseRank and AWP rank wallets differently.
- Evidence: Section 4.4; Table 4.5, whose last row gives the inter-method τ_b with its interval, also with the wallets outside each graph kept as isolated nodes; Figures 4.3 and 4.2.
- Interpretation: the rankings are not equivalent, and their agreement comes from the wallets

that EndorseRank scores zero; the O1 criterion is met (Table 5.1).

8.6.2 H2 / RQ2: Same-window alignment

- Statement: in the same window, each score agrees with the checks built from the edges it walks on.
- Evidence: the family means of every score with their CIs (Table 4.3); the same-window coefficients with the wallets outside a graph kept as isolated nodes (Table 4.5).
- Note: the agreement is partly built in, since the proxies come from the same events as the graphs. EndorseRank’s allowance and transfer coefficients have different ceilings (Section 4.3), so the claim says nothing about which of the two it follows more closely, and the isolated-node rule was tried after the results were known. None of these comparisons was fixed in advance. The paired differences between EndorseRank and AWP and their per-proxy results are in Appendix 8.5.

8.6.3 H3 / RQ3: Computational cost

- Statement: at the same n , a score’s computation time follows the number of edges in its graph, so EndorseRank, on the sparsest graph, runs fastest.
- Evidence: Table 4.1 (runtime, SD, memory, iterations, $|V|$, $|E|$); Tables 8.1 and 4.2; Figure 4.1.
- Note: the expanded-pool stages add sampled addresses but no edges, and hence no solver-graph nodes, beyond the matched-cohort graph (Section 4.2.1); the claim is confined to edge count.

8.6.4 RQ4: Temporal holdout

- Statement: with scores fixed at t_1 , the scores predict the new owner–spender approvals and new transfer senders that appear after t_1 and up to t_2 on the spender cohort ($n = 1,335$), with the edge type deciding which label a score ranks well, but no PageRank predicts either label

better than the raw t_1 degree of its own edge type.

- Evidence: Tables 4.10 and 4.11 (Section 4.6); the June–August window, Table 4.16; the spenders that had received a transfer before the freeze, Table 4.12.
- Qualification: both labels are the constructs as observed in a later period (new approvals, new senders), not credit or trading outcomes, and both are counts of arrivals. RQ4 replaced the proposal’s trading-success validation (Section 2.8). Apart from the increment of C-PR ($\lambda = 1$), none of these comparisons was fixed in advance. The differences between EndorseRank and AWP are in Appendix 8.5.

8.6.5 RQ5: Coupled operator

- Statement: under the rule of 14 September, a per-node convex mixture of the raw-amount allowance and transfer layers (C-PR) does not beat both layers walked alone, and the seeded ablation (S-PR) reproduces the transfer walk. The registered replication does not confirm that C-PR improves on the transfer walk alone, as it did in an exploratory comparison on the spring holdout: F1 and F3 hold, F2 fails. With AWP as the comparator, F2 fails by a wider margin.
- Evidence, rule of 14 September: Table 4.11 for the primary part, and Tables 4.3 and 4.4 for the secondary part (Section 4.3.1).
- Evidence, transfer walk alone: the exploratory spring contrast in Table 4.11, and the registered decision (F1–F3, with F4–F6 reported beside it) in Table 4.15 (Section 4.6.2).
- Evidence, AWP: the registered sensitivity analysis in Table 4.15, and the spring contrasts computed after the labels were known in Table 4.11.
- Evidence, cost: runtime rows of Table 4.1.
- Qualification: tested with fixed $\lambda \in \{0.25, 0.5, 0.75\}$, raw-amount layers, uniform teleportation, transition-level mixing and one chain. The rule of 14 September and the values of λ were set in the configuration before any coupled score was computed (Section 3.7); the rule

for the comparison with the transfer walk alone was written on 28 September 2026 for the June–August window and registered on 1 October 2026 together with two sensitivity analyses, one with AWP as the comparator, before the extraction (Section 3.7.2). The registration documents call the two layers walked alone EndorseRank and AWP.

8.6.6 Claims C1–C5

- C1: the runtime evidence listed under H3/RQ3; the profiling run and the edge ratio are in Section 4.2 and Appendix 8.3.3. The claim is specific to EndorseRank, since C-PR costs at least as much as AWP.
- C2: the same-window agreement of every score with the checks of its own edges (Table 4.3), partly built in and, for EndorseRank, set by the rule for wallets outside the graph (Table 4.5).
- C3: the rank divergence listed under H1/RQ1 (Table 4.5, Figure 4.3).
- C4: the comparison of every PageRank with the raw degree of its own edge type on the spender holdout (Tables 4.10 and 4.11), with the coverage check (Table 4.12), the exploratory trader run as its limit on the traders (Table 4.13) and the June–August window as its repetition (Table 4.16); the robustness checks and the sample-definition sensitivity are reported in Tables 4.6–4.8, Table 4.9 and Figure 4.4.
- C5: coupling does not beat its two layers walked alone, a bounded negative result, and the registered replication does not confirm its exploratory gain over the transfer walk alone (Table 4.15); see RQ5 above and Section 4.3.1.

8.6.7 Explicit non-claims

This research does *not* claim:

- that any of the scores is the better measure of reputation, or that AWP has become obsolete (the comparisons between EndorseRank and AWP in Appendix 8.5 follow the constructs and depend on coverage);

- that EndorseRank aligns more closely with the allowance family than with the transfer family (the two coefficients have different ceilings);
- that any PageRank of this thesis forecasts the future growth of its own construct better than the raw t_1 degree does;
- that the value parameter $b = 1$, or EndorseRank’s uniform restart, is the best choice (both were fixed by this study, the restart after comparing the two rules on these data);
- that C-PR improves on a walk over the transfers alone (the registered rule failed on new senders) or on AWP (it falls behind AWP on new senders in both windows);
- that no combination of the two edge types can beat the single-layer methods, only that the per-node convex mixture tested here, with fixed λ , does not;
- that runtime scales favorably when the edge population grows beyond the cohort graph;
- that, without replication, the results generalize beyond Arbitrum One and the study window;
- that on-chain scores substitute for collateral or legal identity systems.

8.6.8 Replication artifacts

1. A merged wallet-ranking table with the scores of EndorseRank, AWP, C-PR ($\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$) and S-PR.
2. The machine-readable holdout summaries for the spender cohort and the matched traders, with the paired contrasts and primary-criterion verdict (committed under `data/3-published-results-for-thesis/spring-holdout-2026-03-to-2026-05/`).
3. The machine-readable evaluation summary: authoritative same-window numbers, per-run timings and the bootstrap protocol (committed under `data/3-published-results-for-thesis/`).
4. The registered replication: the registration file and plan as committed on 1 October 2026, the extraction manifest with the hashes of both files, the evaluation summary with every coefficient, contrast and verdict, and the post hoc scores of EndorseRank on the same cohorts

- (committed under config-copies/ and registered-replication-2026-06-to-2026-08/ in data/3-published-results-for-thesis/).
5. The exported table fragments (results/tables/).
 6. The exported vector figures (Figures/generated/).
 7. The extracted logs and the tables built from them (data/1-raw-blockchain-logs/ and data/2-processed-tables-and-evaluations/); files over 100 MiB are left out and can be rebuilt from the monthly files.
 8. The archived extended-baseline matrices (data/4-archived-extended-baselines/, optional).

The tables and figures are rebuilt by running the evaluation pipeline on real data with the robustness sweeps, exporting the tables, and then redrawing the figures from the same artifacts (Appendix 8.1.1).

8.7 Ethics Clearance Certificate

8.7.1 Certificate

This study uses public blockchain data only and involves no human participants (Section 3.16). The ethics clearance certificate issued by the Unisa School of Computing / College of Agriculture and Environmental Sciences research ethics review committee is reproduced below.

Ethics clearance certificate Reference: [Unisa SoC/CAES ethics clearance ref: TBD] Date of issue: [TBD] (Scanned certificate to be inserted here.)
--